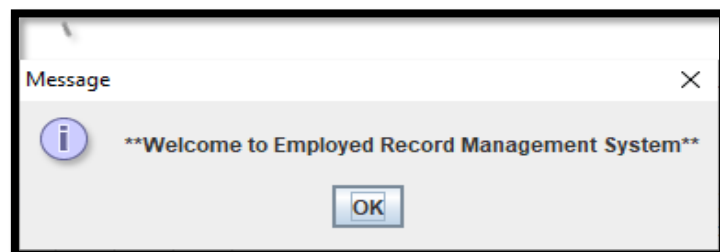


Data structures

Project:

****Employee Record Management System****



=

Student: Mayasim Mohammed Badhurays

Methods:

1-Insert Employee Record:

```
//*****
public void insert_Reecord(int ID,String name,String first_day,
    String phone_number,String address,double worke_hours,double salary){

    Node new_Employee=new Node(ID,name,first_day,phone_number,address,worke_hours,salary);
    if(head==null||head.ID_Employee>new_Employee.ID_Employee){
        new_Employee.next=head;
        head=new_Employee;
        return;
    }

    Node temp=head;
    while(temp.next!=null&&temp.next.ID_Employee<new_Employee.ID_Employee){
        temp=temp.next;
    }
    new_Employee.next=temp.next;
    temp.next=new_Employee;

}

} //End insert_Reecord Method
```

Insert method in main :

```
case 1: //Insert Employee Record

    input = JOptionPane.showInputDialog("Enter your ID ,in range 150 to 250");
    int id = Integer.parseInt(input);
    //validation for id
    while(!flag){
        if(id>=150&&id<=250){
            flag=true;
        }else{
            input= JOptionPane.showInputDialog("wrong ,agin enter ID in range 150 to 250");
            id = Integer.parseInt(input);
        }
    }
    if( !project.cheack_Record(id)){ //
        String name = JOptionPane.showInputDialog("Enter your name");
        String phone = JOptionPane.showInputDialog("Enter your phone");
        input = JOptionPane.showInputDialog("Enter your Salary");
        double salary = Integer.parseInt(input);
        double work_hours;
```

```
        input = JOptionPane.showInputDialog("Enter your work_hours\n"
            + "\" grether than or equale 32 hourse\"");
        work_hours = Double.parseDouble(input);

        flag=false;
        while(!flag){ //validation Work Hourse
            if(work_hours>32.0||work_hours==32.0){
                flag=true;
            }else{
                input = JOptionPane.showInputDialog("wrong,Enter your work hours again\n"
                    + "\" grether than or equale 32 hourse\"");
                work_hours = Double.parseDouble(input);
            }
        }
    }
}
```

```
String address = JOptionPane.showInputDialog("Enter your address");
String First_day = JOptionPane.showInputDialog( "Enter your first day:");
//creat record
project.insert_Reecord(id,name,First_day,phone,address,work_hours,salary);
project.updateSalary(id);
JOptionPane.showMessageDialog(null,"Record Inserted Successfully");

}else{
    JOptionPane.showMessageDialog(null, "Employee already exist ");
}
break;
```

Output of Insert:

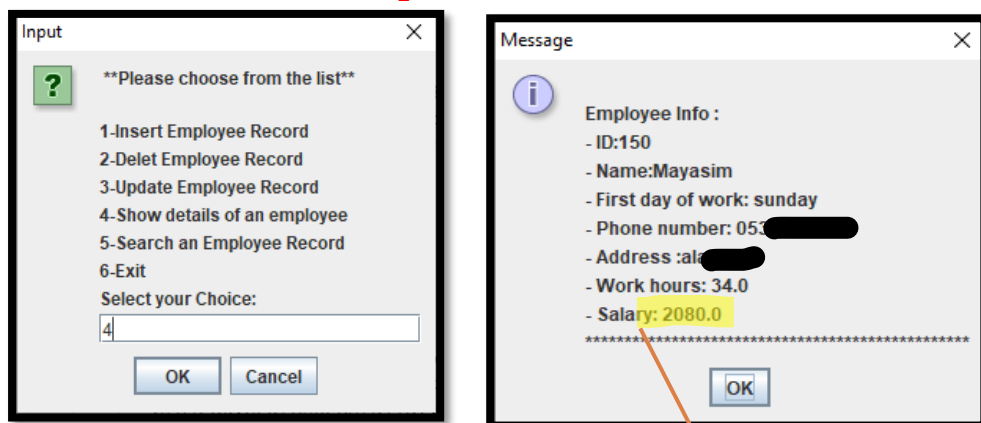
The sequence of dialog boxes is as follows:

- Input Dialog 1:** Title "Input", icon "?", text "Enter your ID, in range 150 to 250", input field contains "149", buttons "OK" and "Cancel".
- Input Dialog 2:** Title "Input", icon "?", text "wrong ,agin enter ID in range 150 to 250", input field contains "150", buttons "OK" and "Cancel".
- Input Dialog 3:** Title "Input", icon "?", text "Enter your name", input field contains "Mayasim", buttons "OK" and "Cancel".
- Input Dialog 4:** Title "Input", icon "?", text "Enter your phone", input field contains "053...", buttons "OK" and "Cancel".
- Input Dialog 5:** Title "Input", icon "?", text "Enter your work_hours", text "grether than or equale 32 hourse", input field contains "21", buttons "OK" and "Cancel".
- Input Dialog 6:** Title "Input", icon "?", text "wrong,Enter your work hours again", text "grether than or equale 32 hourse", input field contains "34", buttons "OK" and "Cancel".
- Input Dialog 7:** Title "Input", icon "?", text "Enter your address", input field contains "...", buttons "OK" and "Cancel".
- Input Dialog 8:** Title "Input", icon "?", text "Enter your first day:", input field contains "sunday", buttons "OK" and "Cancel".
- Message Dialog:** Title "Message", icon "i", text "Record Inserted Successfully", button "OK".

2-Show Of An Employee Record:

```
// *****  
public void showRecord() {  
    Node temp=head;  
    String output="";  
    if(temp==null){  
        JOptionPane.showMessageDialog(null, "No records available..!");  
        return;  
    }  
    while(temp!=null){  
        String data="Employee Info :\n- ID:"+temp.ID_Employee+  
            "\n- Name:"+temp.name_Employee+  
            "\n- First day of work: "+temp.First_day  
            +"\n- Phone number: "+temp.phone_Number+  
            "\n- Address :"+temp.address+  
            "\n- Work hours: "+temp.Work_hours  
            +"\n- Salary: "+temp.salary+  
            "\n*****\n";  
  
        temp=temp.next;  
        output=output+"\n"+data;  
    }  
    JOptionPane.showMessageDialog(null, output );  
} // End showRecord Method
```

Output of Show:

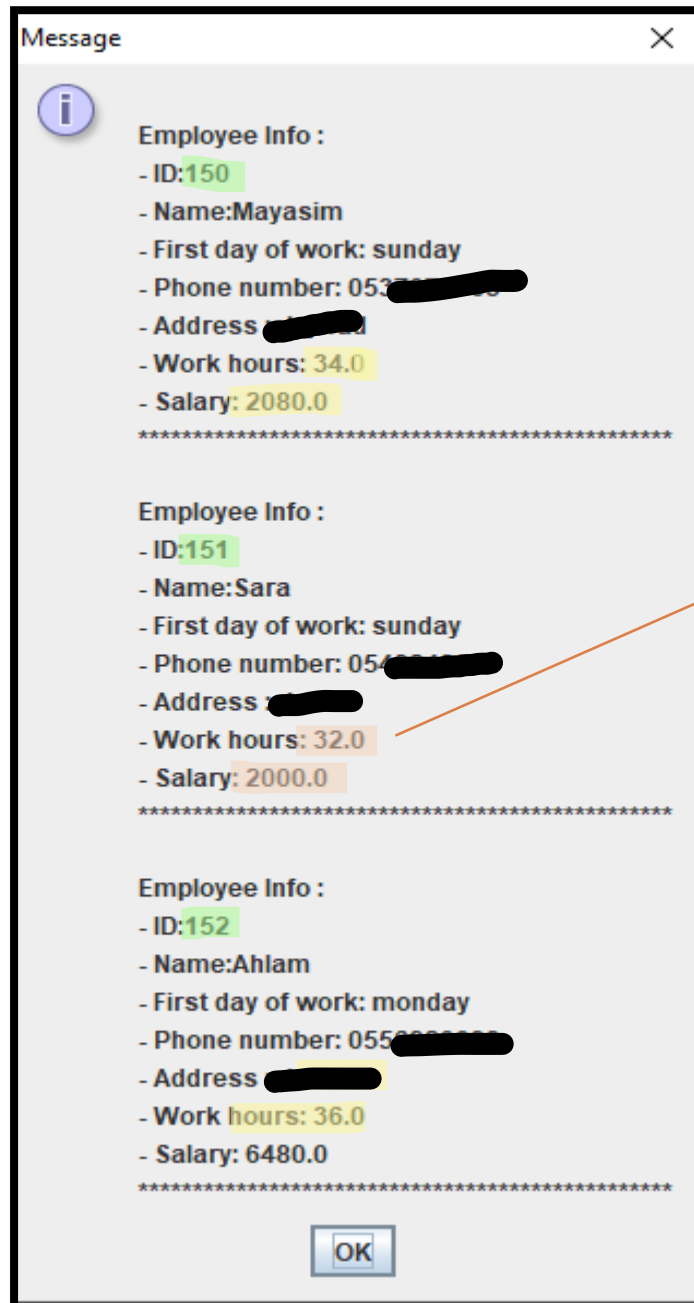


The Salary updated automatically.

Before:2000

After:2080.0 ,when 34 work hours

In show Method the employee record should be inserted in sorted order



No update salary because the number of work hours 32, must be greater than 32 to update

3-Delet Employee Record:

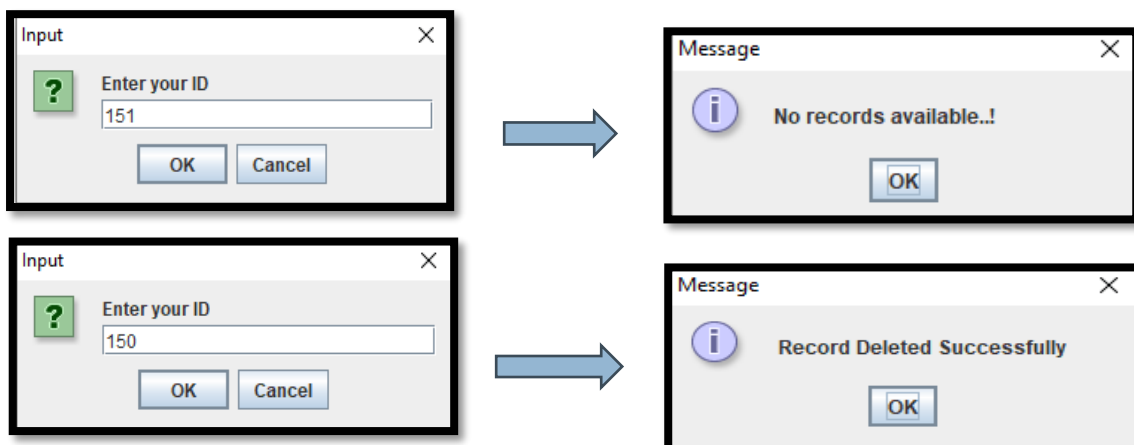
```
//*****
public int delet_Record(int ID) {

    if(!cheack_Record(ID)) {
        JOptionPane.showMessageDialog(null,"No records available..!");
    }
    Node temp=head,tail=null;
    if(temp!=null&&temp.ID_Employee==ID) {
        head=head.next;
        return 0;
    }
    while (temp!=null&&temp.ID_Employee!=ID)
    {
        tail=temp;
        temp=temp.next;
    }
    if(temp==null) {
        return -1;
    }
    tail.next=temp.next;

    return 0;

} //End delet_Record Method
```

Output of Delet Method:



4-Chek Employee Record: use in all operation record

```
//*****
public boolean cheack_Record(int id) {
    Node temp=head;
    if(head==null) {
        return false;
    }
    while(temp!=null) {
        if(temp.ID_Employee==id) {
            return true;
        }
        temp=temp.next;
    } return false;

} //End cheack_Record Method
```

5-Update Employee Record:

```
//*****
public void UpdateRecord(int id){
    if(!check_Record(id)){
        JOptionPane.showMessageDialog(null,"No records available..!");
    }else {

        Node temp = head;
        while (temp!=null){
            if(temp.ID_Employee == id){
                char character = 0;
                do{
                    String list=JOptionPane.showInputDialog("Choose what you want to update:\n"
                        + "1-Name\n"
                        + "2-phone Number\n"
                        + "3-work Hourse\n"
                        + "4-salary\n"
                        + "5-address\n"
                        + "6-First day of work\n"
                        + "Enter your choice:" );

                    int choice= Integer.parseInt(list);

                    switch(choice){

                        case 1:
                            String input = JOptionPane.showInputDialog("enter your name");
                            temp.name_Employee = input;
                            break;

                        case 2:
                            input=  JOptionPane.showInputDialog("Enter your phone");
                            temp.phone_Number = input;
                            break;

                        case 3:
                            input=  JOptionPane.showInputDialog("Enter your work hourse\n"
                                + "\" grether than or equale 32 hourse\"");
                            double number = Double.parseDouble(input);
                            temp.Work_hours =number;
                            updateSalary(id);
                            break;

                        case 4:
                            input = JOptionPane.showInputDialog(" Enter your salary");
                            number= Double.parseDouble(input);
                            temp.salary=number;
                            break;

                        case 5:
                            input =JOptionPane.showInputDialog("Enter your address: ");
                            temp.address = input;
                            break;

                        case 6:
                            input=JOptionPane.showInputDialog("Enter your first day: ");
                            temp.First_day = input;
                            break;

                        default:
                            JOptionPane.showMessageDialog(null,"Error selection");
                            list=JOptionPane.showInputDialog( "Do you want to reselect?Y/N");
                            character=list.charAt(0);
                            while(character=='Y' || character=='y');

                    }

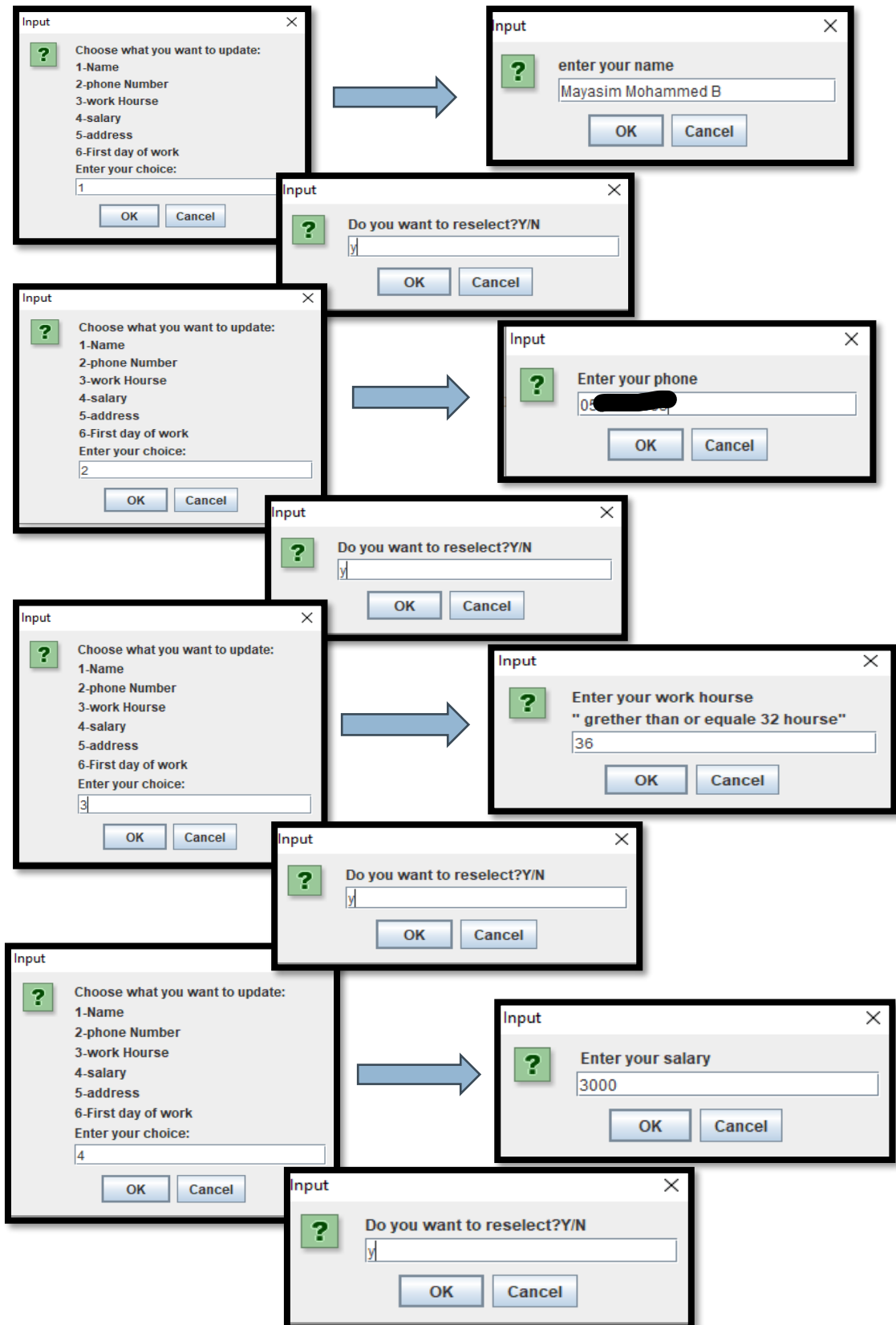
                    temp=temp.next;

                }

            }
        }
    }
}

}
} // End UpdateRecord Method
```

Output Of Update Method:



Input

Choose what you want to update:

- 1-Name
- 2-phone Number
- 3-work Hourse
- 4-salary
- 5-address
- 6-First day of work

Enter your choice:

5

OK Cancel



Input

Enter your address:

██████████

OK Cancel

Input

Do you want to reselect?Y/N

y

OK Cancel

Input

Choose what you want to update:

- 1-Name
- 2-phone Number
- 3-work Hourse
- 4-salary
- 5-address
- 6-First day of work

Enter your choice:

6

OK Cancel



Input

Enter your first day:

Monday

OK Cancel

Input

Do you want to reselect?Y/N

y

OK Cancel

Input

Choose what you want to update:

- 1-Name
- 2-phone Number
- 3-work Hourse
- 4-salary
- 5-address
- 6-First day of work

Enter your choice:

7

OK Cancel



Message

Error selection

OK

Input

Do you want to reselect?Y/N

n

OK Cancel

Exist from update

Re-shows the operations menu.

Input

Do you want another operation? if yes press Y if No press N

y

OK Cancel

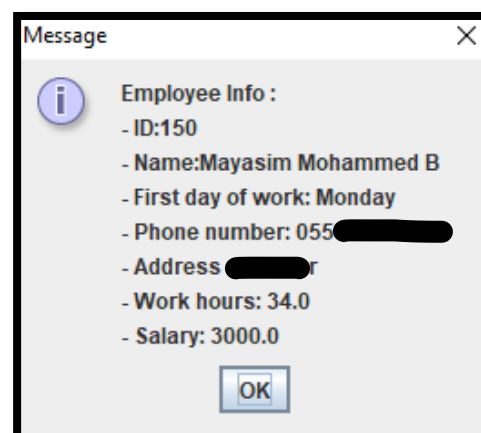
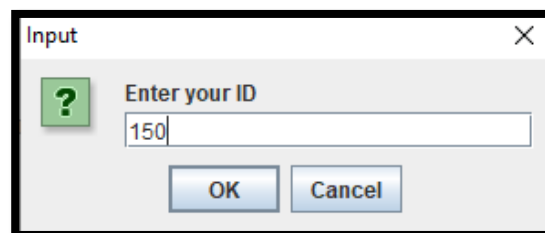
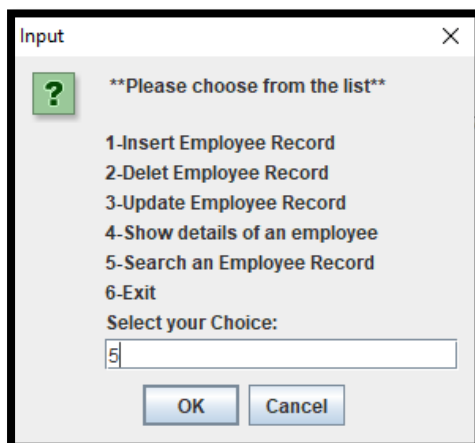
6-Search An Employee:

```

//*****
public void SearchRecord(int id){
    if(!cheack_Record(id)){
        JOptionPane.showMessageDialog(null,"No records available...!");
    }else {
        Node temp = head;
        while (temp!=null){
            if(temp.ID_Employee == id){
                JOptionPane.showMessageDialog(null,"Employee Info :\n- ID:"+temp.ID_Employee
                    +"\n- Name:"+temp.name_Employee
                    +"\n- First day of work: "+temp.First_day
                    +"\n- Phone number: "+temp.phone_Number
                    +"\n- Address :"+temp.address
                    +"\n- Work hours: "+temp.Work_hours
                    +"\n- Salary: "+temp.salary);
                break;
            }
            temp=temp.next;
        }
    }
}
//End SearchRecord Method

```

Output Of Search Method:



7-Update The Salary Of An Employee:

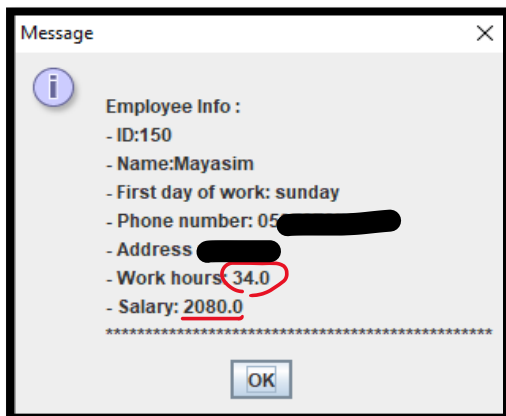
```
//*****  
public void updateSalary(int id) {  
    Node temp = head;  
  
    while (temp != null) {  
        if (temp.ID_Employee == id) {  
            if (temp.Work_hours > 32) {  
                double hours = temp.Work_hours - 32;  
                temp.salary += (temp.salary * 0.02) * hours;  
            }  
        }  
        temp = temp.next;  
    }  
}  
}  
//End Updat Salary Method
```

Output Of Updat Salary Method:

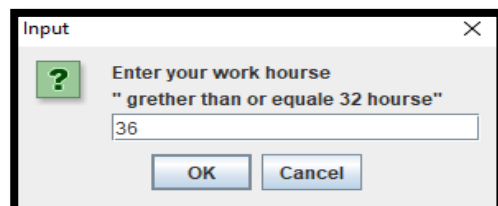
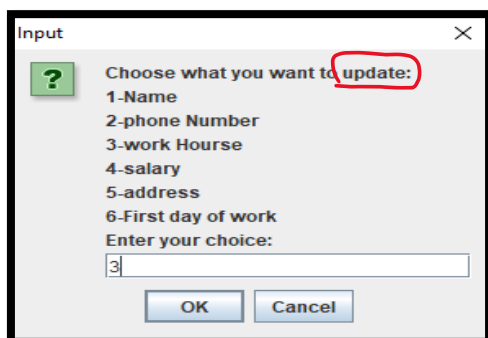
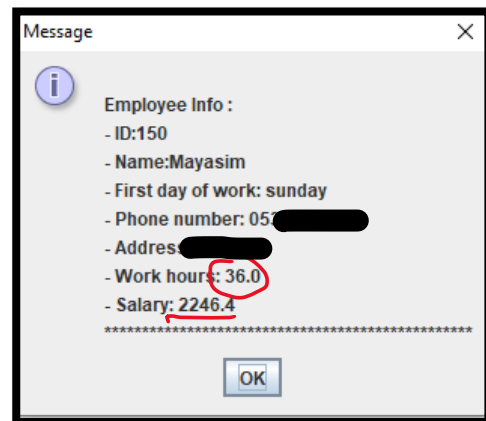
Example 1: in Show Method after insert record .

Example 2: when Update Work Hours to 36 :

Before:



After:



9- Main Method:

```

//*****
public static void main(String[] args) {
    // TODO code application logic here
    Project_Data_Structures project=new Project_Data_Structures ();
    JOptionPane.showMessageDialog(null,"**Welcome to Employed Record Management System**\n" );
    char character;
    String input;

    do{
        String list= JOptionPane.showInputDialog( " **Please choose from the list** \n\n"
            + "1-Insert Employee Record\n"
            + "2-Delet Employee Record\n"
            + "3-Update Employee Record\n"
            + "4-Show details of an employee\n"
            + "5-Search an Employee Record\n"
            + "6-Exit\n"
            + "Select your Choice:");

        int choice= Integer.parseInt(list);
        boolean flag=false;

        switch(choice){

            case 1: //Insert Employee Record

                input =  JOptionPane.showInputDialog("Enter your ID ,in range 150 to 250");
                int id = Integer.parseInt(input);

```

```

//validation for id
while(!flag){
    if(id>=150&&id<=250){
        flag=true;
    }else{
        input= JOptionPane.showInputDialog("wrong ,agin enter ID in range 150 to 250");
        id = Integer.parseInt(input);
    }
}
if( !project.cheack_Record(id)){ //
String name = JOptionPane.showInputDialog("Enter your name");
String phone =  JOptionPane.showInputDialog("Enter your phone");
input =  JOptionPane.showInputDialog("Enter your Salary");
double salary = Integer.parseInt(input);
double work_hours;

input =  JOptionPane.showInputDialog("Enter your work_hours\n"
    + "\" grether than or equale 32 hourse\"");
work_hours = Double.parseDouble(input);

flag=false;
while(!flag){//validation Work Hourse
    if(work_hours>32.0||work_hours==32.0){
        flag=true;
    }else{
        input =  JOptionPane.showInputDialog("wrong,Enter your work hours again\n"

```

```

    }else{
        input = JOptionPane.showInputDialog("wrong,Enter your work hours again\n"
            + "\n grether than or equale 32 hourse\n");
        work_hours = Double.parseDouble(input);
    }

}

String address = JOptionPane.showInputDialog("Enter your address");
String First_day = JOptionPane.showInputDialog( "Enter your first day:");
//creat record
project.insert_Reecord(id,name,First_day,phone,address,work_hours,salary);
project.updateSalary(id);
JOptionPane.showMessageDialog(null,"Record Inserted Successfully");

}else{
    JOptionPane.showMessageDialog(null, "Employee already exist ");
}
break;

```

```

case 2:
    input = JOptionPane.showInputDialog("Enter your ID");
    id = Integer.parseInt(input);
    if(project.delet_Record(id)==0) {
        JOptionPane.showMessageDialog(null, " Record Deleted Successfully\n");
    }

    break;

case 3:

    input = JOptionPane.showInputDialog("Enter your ID ");
    id = Integer.parseInt(input);
    project.UpdateRecord(id);
    break;

case 4:
    project.showRecord();
    break;

case 5:
    input = JOptionPane.showInputDialog("Enter your ID");
    id = Integer.parseInt(input);
    project.SearchRecord(id);
    break;

```

```

case 6:
    JOptionPane.showMessageDialog(null, "Thank you for using the system ");
    System.exit(0);

default:
    JOptionPane.showMessageDialog(null, " error, please select again");
    list= JOptionPane.showInputDialog( " **Please choose from the list** \n\n"
        + "1-Insert Employee Record\n"
        + "2-Delet Employee Record\n"
        + "3-Update Employee Record\n"
        + "4-Show details of an employee\n"
        + "5-Search an Employee Record\n"
        + "6-Exit\n"
        + "Select your Choice:");
    break;
}

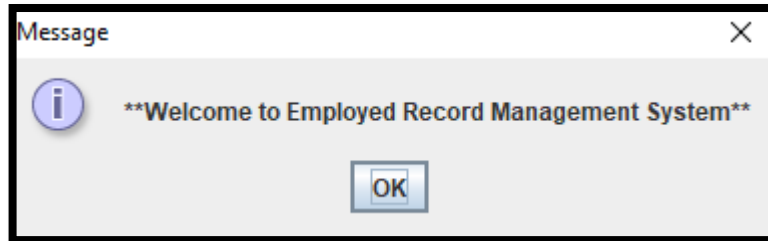
input=JOptionPane.showInputDialog("Do you want another operation? if yes press Y if No press N");
character=input.charAt(0);
}while(character=='Y' || character=='y');

} //End main method

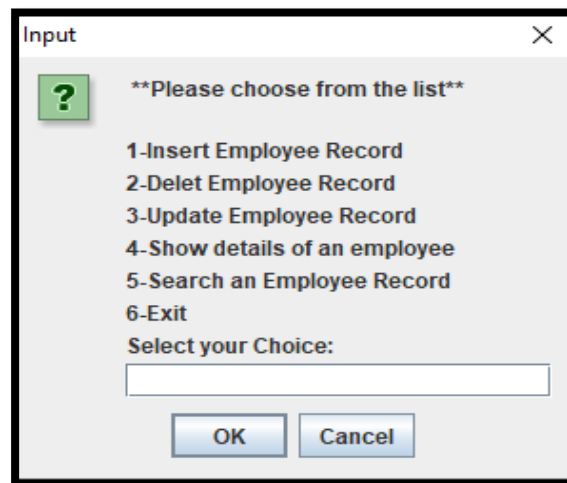
```

Output Of MainMethod:

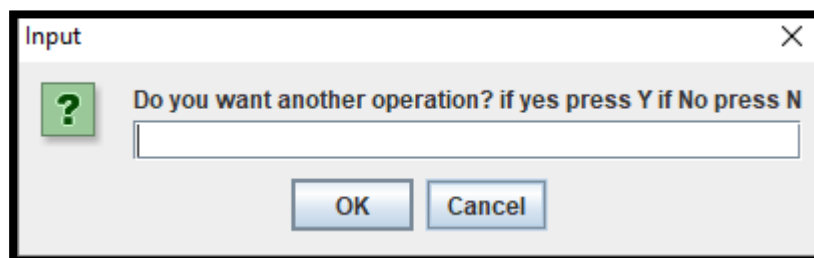
1- Welcome, Message:



2- Menu Operation of employee record:

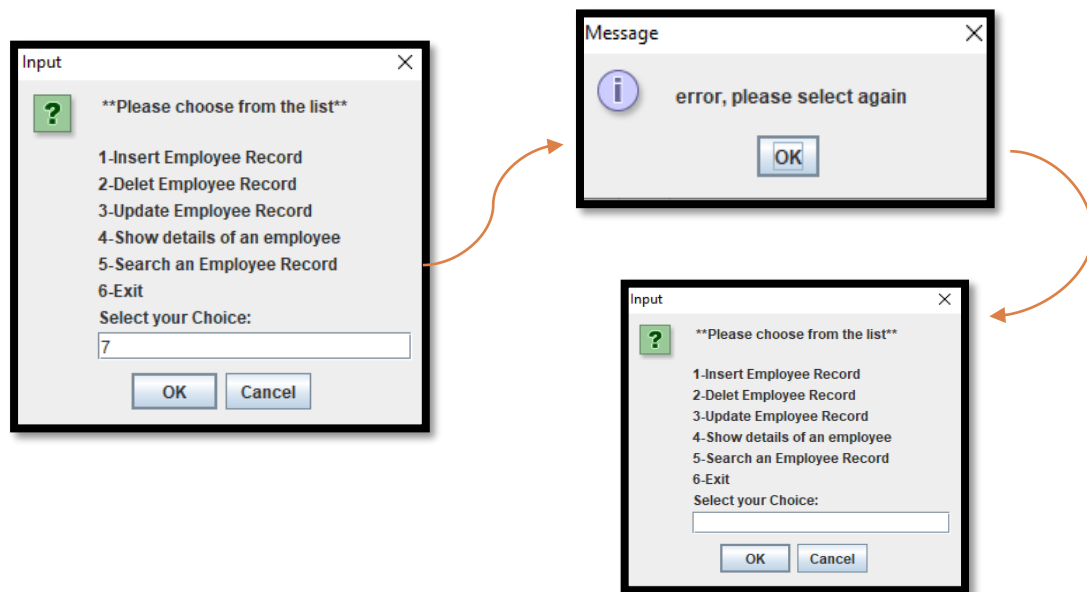


4- After each operation, this message appears:

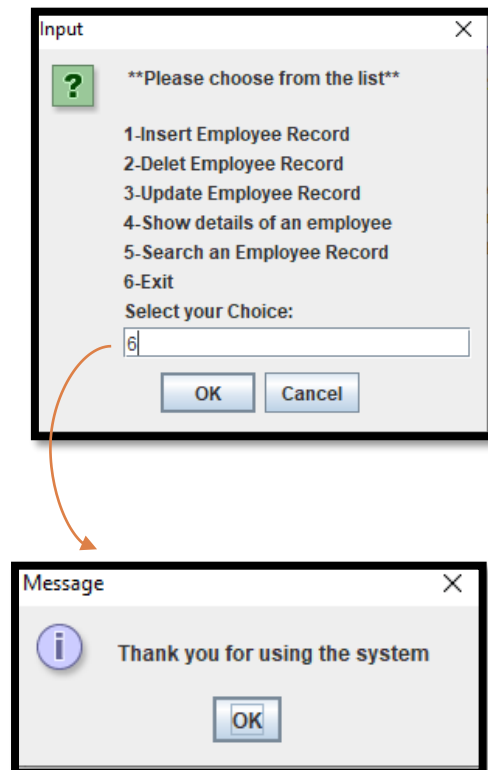


If he chooses y, he returns the menu, or if he chooses n, he exits the program

5-if choice any number outside the menu ,Error message appears and return the menu



6- if choice 6, A thank you message appears and then exits the program:



9-Class and Constructor method of Node:

```
package project_data_structures;

import javax.swing.*;
import java.util.*;

public class Project_Data_Structures {
    private static Node head;

    /**
     *
     */
    private class Node{
        private int ID_Employee;
        private String name_Employee,phone_Number,address,First_day;
        private double Work_hours,salary;
        private Node next;

        //here constructor of class Node
        public Node(int ID,String name,String day,String phone_Number,
                    String address,double hours,double salary){
            this.ID_Employee=ID;
            this.name_Employee=name;
            this.First_day=day;
            this.phone_Number=phone_Number;
            this.address=address;
            this.Work_hours=hours;
            this.salary=salary;
            this.next=null;
        }
    }
}

// class Node
```